Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A conducting sphere of radius R is given a charge
Q. The electric potential and the electric field at the centre of the sphere respectively are:

(A) 0 and $\frac{\mathrm{Q}}{4\pi\varepsilon_0\mathrm{R}^2}$

(B) $\frac{\mathrm{Q}}{4\pi\varepsilon_0\mathrm{R}}$ and 0

(C) $\frac{\mathrm{Q}}{4\pi\varepsilon_0\mathrm{R}}$ and $\frac{\mathrm{Q}}{4\pi\varepsilon_0\mathrm{R}^2}$

(D) Both are 0

Q.2

Voltage rating of a parallel plate capacitor is $500\ \mathrm{V}$. Its dielectric can withstand a maximum electric field of $10^6\ \mathrm{V\ m^{-1}}$. The plate area is $10^{-4}\ \mathrm{m^2}$. What is the dielectric constant if the capacitance is $15\mathrm{pF}$? (given $\epsilon 0 = 8.86 \times 10^{-12}\mathrm{C^2\ N^{-1}\ m^{-2}}$)

(A) 3.8

(B) 8.5

(C) 6.2

(D) 4.5

Q.3

The expression $\mathrm{E} = -\frac{\mathrm{dv}}{\mathrm{dr}}$ implies, that electric field is in that direction in which

(A) increase in potential is steepest.

(B) decrease in potential is steepest.

(C) change in potential is minimum.

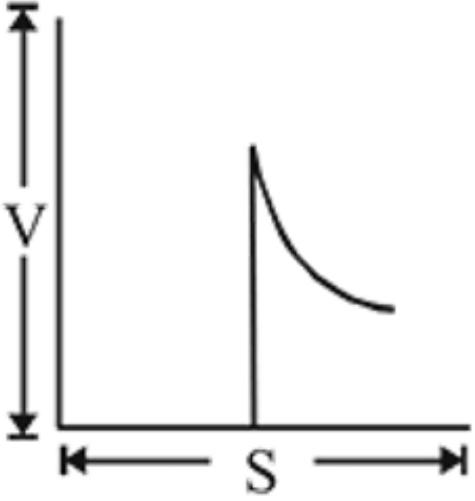
(D) None of these

Q.4

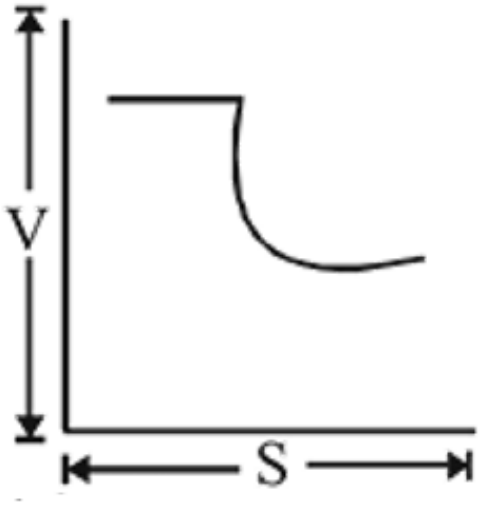
In a hollow spherical shell, potential (V) changes with respect to distance (s) from centre as

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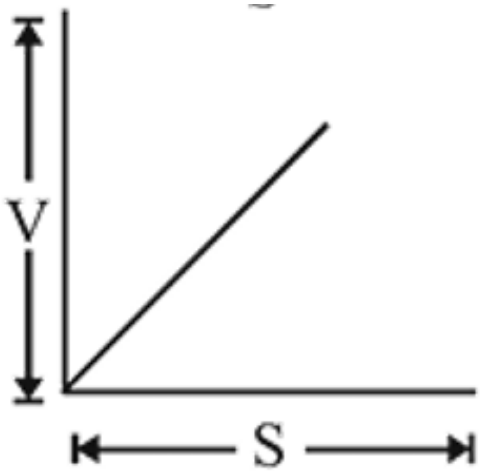
(A)



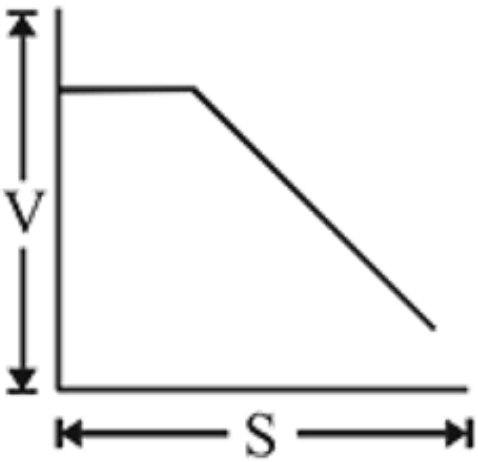
(B)



(C)



(D)



Q.5

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The 1000 small droplets of water each of radius and charge Q, make a big drop of spherical shape. The potential of big drop is how many times the potential of one small droplet?

(A) 1

(B) 10

(C) 100

(D) 1000

Q.6

The electric potential at a point (x, y) in the $x-y$ plane is given $\mathrm{by} \mathrm{V}=-\mathrm{kxy}$. The field intensity at a distance r from the origin varies as

(A) r^2

(B) \(\frac{1}{{\mathrm{r}^{2}}}\)

(C) $1/\mathrm{r}$

(D) r

Q.7

Four charges $\mathrm{q}_1=2 \times 10^{-8} \mathrm{C}, \mathrm{q}_2=-2 \times 10^{-8} \mathrm{C}, \mathrm{q}_3$
$=-3 \times 10^{-8} \mathrm{C}$, and $\mathrm{q}_4=6 \times 10^{-8} \mathrm{C}$ are placed at
four corners of a square of side $\sqrt{2}\ \mathrm{m}$. What is the potential at the centre of the square?

(A) $270\ \mathrm{V}$

(B) $300\ \mathrm{V}$

(C) Zero

(D) $100\ \mathrm{V}$

Q.8

A charge Q is distributed over two concentric hollow spheres of radii r and $R(R>r)$ such that the surface densities are equal. The potential at the common centre is $\frac{1}{4 \pi \varepsilon_0}$ times-

(A) $Q\left[\frac{r+R}{r^2+R^2}\right]$

(B) $\frac{Q}{2}\left(\frac{r+R}{r^2+R^2}\right)$

(C) $2Q\left(\frac{r+R}{r^2+R^2}\right)$

(D) zero

Q.9

Three concentric charged metallic spherical shells A, B and C have radii a, b and c; charge densities $\sigma, -\sigma$

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and σ and potentials V_A, V_B and V_C respectively. Then which of the following relations is correct?

(A) $V_A = (a+b+c)\frac{\sigma}{\varepsilon_0}$

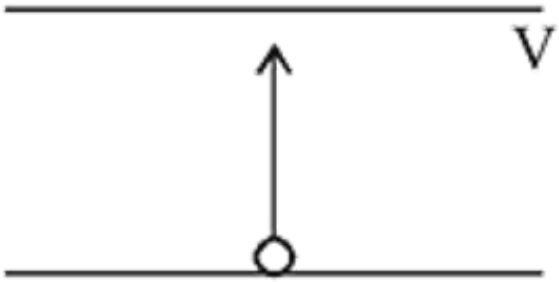
(B) $V_B = \left(\frac{a^2}{b} - b + c\right)\frac{\sigma}{\varepsilon_0}$

(C) $V_C = \left(\frac{a^2+b^2}{b} + c\right)\frac{\sigma}{\varepsilon_0}$

(D) $V_A = V_B = V_C = (a+b+c)\frac{\sigma}{\varepsilon_0}$

Q.10

An electron having charge e and mass m starts from the lower plate of two metallic plates separated by a distance d. If the potential difference between the plates is V, the time taken by the electron to reach the upper plate is given by



(A) $\sqrt{\frac{2md^2}{eV}}$

(B) $\sqrt{\frac{md^2}{eV}}$

(C) $\sqrt{\frac{md^2}{2eV}}$

(D) $\frac{2md^2}{eV}$

Q.11

There is a uniform electrostatic field in a region. The potential at various points on a small sphere centred at P, in the region, is found to vary between in the limits $589.0\ \mathrm{V}$ to $589.8\ \mathrm{V}$. What is the potential at a point on the sphere whose radius vector makes an angle of 60° with the direction of the field ?

(A) $589.5\ \mathrm{V}$

(B) $589.2\ \mathrm{V}$

(C) $589.4\ \mathrm{V}$

(D) $589.6\ \mathrm{V}$

Q.12

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A charge of 3 coulomb moving in a uniform electric field experiences a force of 3000 newton. The potential difference between the two points situated in a field at a distance of $1\ \mathrm{cm}$ from each other will be :

(A) 100

(B) 5000

(C) 10

(D) 50

Q.13

Three charges $-q, +q$ and $+q$ are situated in $X-Y$ plane at points $(0,-a),(0,0)$ and $(0, a)$ respectively. The potential at a point distant $\mathrm{r}(r>a)$ in a direction making an angle θ from Y -axis will be

(A) $\frac{\mathrm{K}q}{r}\left(1-\frac{2a\cos\theta}{r}\right)$

(B) $\frac{2\mathrm{k}q\cos\theta}{r^2}$

(C) $\frac{\mathrm{K}q}{r}$

(D) $\frac{\mathrm{K}q}{r}\left(1+\frac{2a\cos\theta}{r}\right)$

Q.14

In a region, the potential is represented by $\mathrm{V}(\mathrm{x}, \mathrm{y}, \mathrm{z})=6\mathrm{x}-8\mathrm{xy}-8\mathrm{y}+6\mathrm{yz}$, where V is in volts and $\mathrm{x}, \mathrm{y}, \mathrm{z}$ are in metres. The electric force experienced by a charge of 2 coulomb situated at $\mathrm{point}(1,1,1)$ is :

(A) $6\sqrt{5}\ \mathrm{N}$

(B) $30\ \mathrm{N}$

(C) $24\ \mathrm{N}$

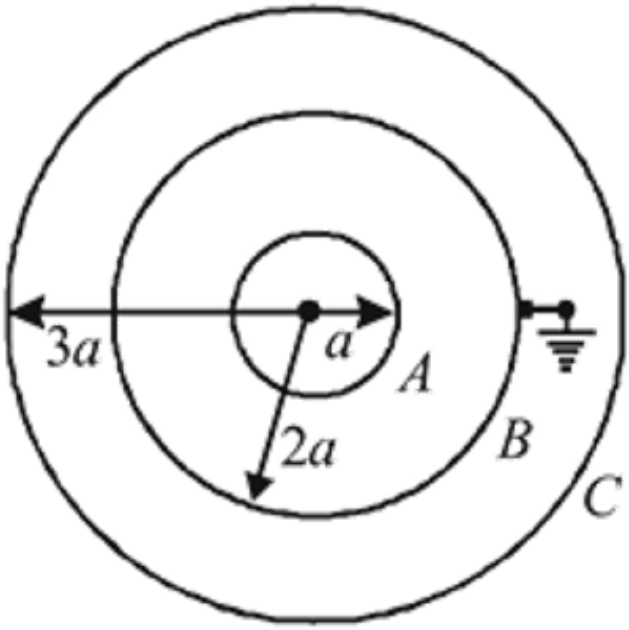
(D) $4\sqrt{35}\ \mathrm{N}$

Q.15

Figure shows a system of three concentric metal shells A, B and C with radii $a, 2a$ and $3a$ respectively. Shell B is earthed and shell C is given a charge Q. Now if shell C is connected to shell A, then the final charge on the shell B, is

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equal to



(A) $-4Q/13$

(B) $-8Q/11$

(C) $-5Q/3$

(D) $-3Q/7$

Q.16

A charge $+q$ fixed at each of the points $x = x_0, x = 3x_0, x = 5x_0, \ldots$ upto ∞ on X -axis and charge $-q$ is fixed on each of the points $x = 2x_0, x = 4x_0$, upto ∞. Here x_0 is a positive constant. Take the potential at a point due to a charge Q at a distance r form it to be $\frac{Q}{4\pi\varepsilon_0 r}$, then the potential at the origin due to above system of charges will be:

(A) zero

(B) infinite

(C) $\frac{q}{8\pi\varepsilon_0 x_0 \log_e 2}$

(D) $\frac{q \log_e 2}{4\pi\varepsilon_0 x_0}$

Q.17

A sphere of radius R has a uniform charge density ρ. The difference in the electric potential at $r = R$ and $r = 0$ from the centre is

(A) $\frac{-\rho R^2}{\epsilon_0}$

(B) $\frac{-2\rho R^2}{\epsilon_0}$

(C) $\frac{\rho R^2}{3\epsilon_n}$

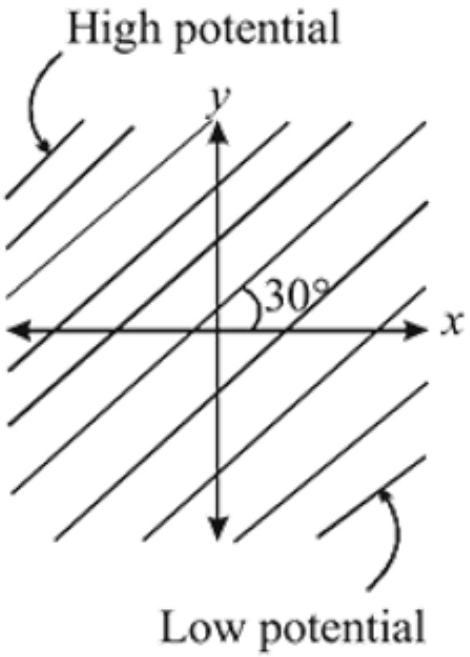
(D) $\frac{-\rho R^2}{6\epsilon_0}$

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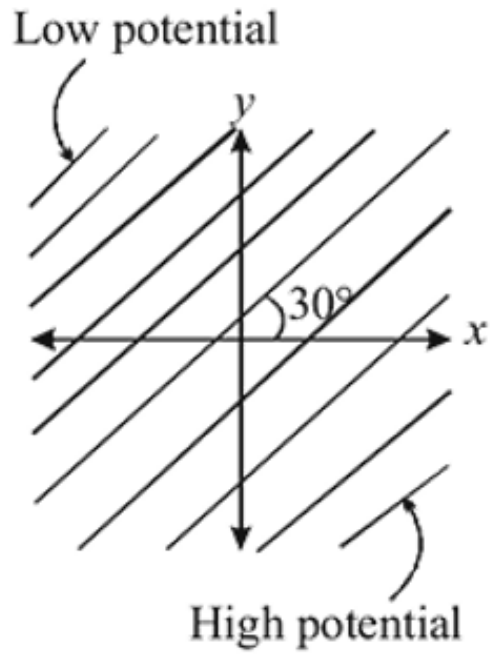
Q.18

The electric field intensity at all points in space
is given by $\vec{E} = \sqrt{3}\hat{i} - \hat{j}$ volt/metre. The nature of equipotential lines in xy -plane is given by

(A)

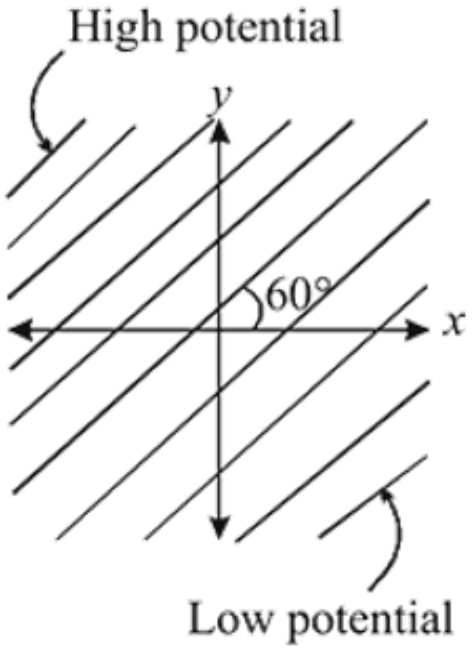


(B)

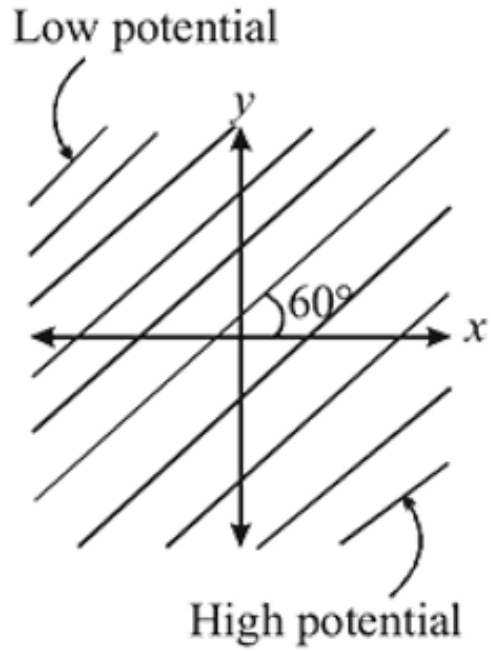


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(C)



(D)



Q.19

The electric potential at a point (x, y, z) is given by $V = -x^2y - xz^3 + 4$. The electric field $\vec{E}$ at that point is

(A) $\vec{E} = \hat{i}2xy + \hat{j}\left(x^2 + y^2\right) + \hat{k}\left(3xz - y^2\right)$

(B) $\vec{E} = \hat{i}z^3 + \hat{j}xyz + \hat{k}z^2$

(C) $\vec{E} = \hat{i}\left(2xy - z^3\right) + \hat{j}xy^2 + \hat{k}3z^2x$

(D) $\vec{E} = \hat{i}\left(2xy + z^3\right) + \hat{j}x^2 + \hat{k}3xz^2$

Q.20

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Electric field at a far away distance r on the axis of dipole is $\vec{E}_0$. What is the electric field at a distance $2r$ on perpendicular bisector?

(A) $\frac{\vec{E}_0}{16}$

(B) $-\frac{\vec{E}_0}{16}$

(C) $\frac{\vec{E}_0}{8}$

(D) $-\frac{\vec{E}_0}{8}$

Answers & Solutions

Q.1 Answer:

$\frac{\mathrm{Q}}{4\pi\varepsilon_0\mathrm{R}}$ and 0

Solution:

Due to conducting sphere At centre, electric field $\mathrm{E} = 0$

And electric potential $\mathrm{V} = \frac{\mathrm{Q}}{4\pi\in_0\mathrm{R}}$

Q.2 Answer:

8.5

Solution:

$$\mathrm{C} = \mathrm{K}\varepsilon_0\ \mathrm{A}/\mathrm{d} \text{ and } \mathrm{V} = \mathrm{Ed}$$
$$\text{Or } \mathrm{K} = \mathrm{CV}/\varepsilon_0\mathrm{AE}_{\max}$$
$$\mathrm{K} = \left(15 \times 10^{-12} \times 500\right) / \left(8.86 \times 10^{-12} \times 10^{-4} \times 10^{6}\right) = 8.5$$

Q.3 Answer:

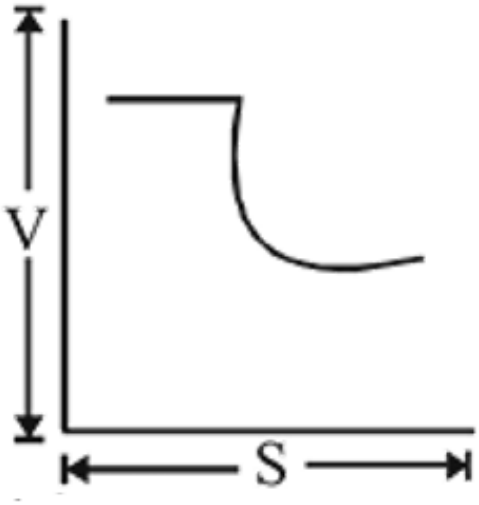
decrease in potential is steepest.

Solution:

No solution available

Q.4 Answer:

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Solution:

No solution available

Q.5 Answer:

100

Solution:

Volume of big drop $=1000\times$ **volume of each small drop** $\frac{4}{3}\pi \mathrm{R}^3=1000\times\frac{4}{3}\pi \mathrm{r}^3 \Rightarrow \mathrm{R}=10\mathrm{r}$

$\because \quad V=\frac{kq}{r}$ **and** $V'=\frac{kq}{R}\times 1000$

Total charge on one small droplet is q and

on the big drop is $1000\mathrm{q}. \Rightarrow \quad \frac{\mathrm{V}'}{\mathrm{V}}=\frac{1000\mathrm{r}}{\mathrm{R}}=\frac{1000}{10}=100$

$\therefore \quad V'=100\ \mathrm{V}$

Q.6 Answer:

r

Solution:

Let the $x-y$ **coordinates of the point at distance** r **from the origin be given as** $x=r\cos\theta, y=r\sin\theta$

Potential is given as $\mathrm{V}=-\mathrm{Kxy}$

Now, Electric field intensity is $\vec{E}=(-\mathrm{dV}/\mathrm{dx})\vec{i}+(-\mathrm{dV}/\mathrm{dy})\vec{j}=\mathrm{K}(\mathrm{y}\vec{i}+\mathrm{x}\vec{j})=\mathrm{K}(r\sin\theta\vec{i}+\mathrm{r}\cos\theta\vec{j})=\mathrm{Kr}(\sin\theta\vec{i}+\cos\theta\vec{j})\propto \mathrm{r}$

So electric field potential is proportional to r.

Q.7 Answer:

270 V

Solution:

Potential at the centre $\mathrm{O}, \mathrm{V}=\mathrm{k}\frac{\mathrm{q}}{\mathrm{r}}$

$\mathrm{V}=\mathrm{k}\left[\frac{2\times10^{-8}}{1}+\frac{-2\times10^{-8}}{1}+\frac{-3\times10^{-8}}{1}+\frac{6\times10^{-8}}{1}\right]$

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$V = k \times 3 \times 10^{-8} = 9 \times 10^9 \times 3 \times 10^{-8}$volt

$= 27 \times 10 = 270$ V

Q.8 Answer:

$Q\left[\frac{r+R}{r^2+R^2}\right]$

Solution:

$q_r + q_R = Q$ **...(1)**

$\frac{q_r}{q_R} = \frac{4\pi r^2\sigma}{4\pi R^2\sigma} = \frac{r^2}{R^2} \Rightarrow \frac{q_r}{q_R} = \frac{r^2}{R^2} \ldots (2)$

From (1) and (2) $q_r = \frac{Qr^2}{R^2+r^2}$ **and** $q_R = \frac{QR^2}{R^2+r^2}$

So, $V = \frac{q_r}{4\pi\varepsilon_0 r} + \frac{q_R}{4\pi\varepsilon_0 R}$

$\Rightarrow V = \frac{Q}{4\pi\varepsilon_0}\left(\frac{r+R}{r^2+R^2}\right)$

Q.9 Answer:

$V_B = \left(\frac{a^2}{b} - b + c\right)\frac{\sigma}{\varepsilon_0}$

Solution:

The three shells are shown in figure.

(i) Potential of A

$= (\text{ Potential of A due to } +\sigma \text{ on A })$

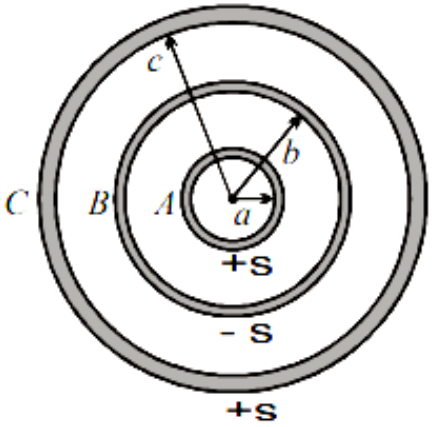
$+ (\text{ Potential of A due to } -\sigma \text{ on B })$

$+ (\text{ Potential of A due to } +\sigma \text{ on C })$

$= K\left[\frac{4\pi a^2\sigma}{a} - \frac{4\pi b^2\sigma}{b} + \frac{4\pi c^2\sigma}{c}\right]$

$= \frac{\sigma}{\epsilon_0}[a - b + c]$

Potential of $B = (\text{ Potential due to } +\sigma \text{ on A}) + (\text{ Potential due to } -\sigma \text{ on B})$



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$+$ (Potential due to $+\sigma$ on C)

$$= \mathrm{K}\left[\frac{4\pi \mathrm{a}^2\sigma}{\mathrm{a}} - \frac{4\pi \mathrm{b}^2\sigma}{\mathrm{b}} + \frac{4\pi \mathrm{c}^2\sigma}{\mathrm{c}}\right]$$

$$= \frac{\sigma}{\epsilon_0}\left[\frac{\mathrm{a}^2}{\mathrm{b}} - \mathrm{b} + \mathrm{c}\right]$$

Potential of $\mathrm{C} =$ (**Potential due to** $+\sigma$ **on** $\mathrm{A}) +$ (**Potential due to** $-\sigma$ **on** $\mathrm{B}) +$ (**Potential due to** $+\sigma$ **on** C)

$$= K\left[\frac{4\pi \mathrm{a}^2\sigma}{\mathrm{a}} - \frac{4\pi \mathrm{b}^2\sigma}{\mathrm{b}} + \frac{4\pi \mathrm{c}^2\sigma}{\mathrm{c}}\right]$$

$$= \frac{\sigma}{\epsilon_0}\left[\frac{\mathrm{a}^2}{\mathrm{b}} - \frac{\mathrm{b}^2}{\mathrm{c}} + \mathrm{c}\right]$$

Q.10 Answer:

$\sqrt{\frac{2\mathrm{md}^2}{\mathrm{eV}}}$

Solution:

$\mathrm{E} = \frac{\mathrm{V}}{\mathrm{d}}, \mathrm{F} = \mathrm{eE} = \mathrm{eV/d}$

$\mathrm{a} = \frac{\mathrm{F}}{\mathrm{m}} = \frac{\mathrm{eV}}{\mathrm{md}}$

$\mathrm{d} = \frac{1}{2}\mathrm{at}^2$ **or** $\mathrm{t} = \sqrt{\frac{2\,\mathrm{d}}{\mathrm{a}}}$

or $\mathrm{t} = \sqrt{\frac{2\,\mathrm{dmd}}{\mathrm{eV}}} = \sqrt{\frac{2\mathrm{md}^2}{\mathrm{eV}}}$

Q.11 Answer:

589.4 V

Solution:

$$\Delta \mathrm{V} = \mathrm{E.d}$$
$$= \mathrm{Ed}\cos\theta = 0.8 \times \cos 60^\circ$$
$$= 0.4$$

Hence the new potential at the point on the sphere is $589.0 + 0.4 = 589.4$ V

Q.12 Answer:

10

Solution:

Force on a charge in an electric field is $F = qE$

But $E = \frac{V}{d} \quad \therefore \quad F = \frac{qV}{d}$

or $\quad V = \frac{Fd}{q} = \frac{300\ \mathrm{N} \times 1 \times 10^{-2}\ \mathrm{m}}{3C} = 10$ **volt**

Q.13 Answer:

$\frac{\mathrm{K}q}{r}\left(1 + \frac{2a\cos\theta}{r}\right)$

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Solution:

$$V = \frac{Kq \cdot 2a\cos\theta}{r^2} + \frac{Kq}{r} = \frac{Kq}{r}\left[\frac{2a\cos\theta}{r} + 1\right]$$

Q.14 Answer:

$4\sqrt{35}\ \mathrm{N}$

Solution:

$$\mathrm{V}_{(x,y,x)} = 6\mathrm{x} - 8\mathrm{xy} - 8\mathrm{y} + 6\mathrm{yz}$$
$$\mathrm{E_x} = -\frac{\partial \mathrm{V}}{\partial \mathrm{x}} = -6 + 8\mathrm{y}$$
$$\mathrm{E_y} = -\frac{\partial \mathrm{V}}{\partial \mathrm{y}} = 8\mathrm{x} + 8 - 6\mathrm{z}$$
$$\mathrm{E}_z = -\frac{\partial \mathrm{V}}{\partial \mathrm{z}} = -6\mathrm{y}$$
$$\overrightarrow{\mathrm{E}} = (-6 + 8\mathrm{y})\hat{\mathrm{i}} + (8\mathrm{x} + 8 - 6\mathrm{z})\hat{\mathrm{j}} - 6\mathrm{y}\hat{\mathrm{k}}$$
$$\mathrm{E}_{(1,1,1)} = 2\hat{\mathrm{i}} + 10\hat{\mathrm{j}} - 6\hat{\mathrm{k}}$$
$$\overrightarrow{|\mathrm{E}|} = 2\sqrt{35}\mathrm{NC}^{-1}$$
$$\mathrm{F} = \mathrm{qE} = 2 \times 2\sqrt{35} = 4\sqrt{35\ \mathrm{N}}$$

Q.15 Answer:

$-8Q/11$

Solution:

From given conditions, $V_A = V_C$ **and** $V_B = 0 \Rightarrow V_B = \frac{k(Q-q_1)}{3a} + \frac{kq_2}{2a} + \frac{kq_1}{2a} = 0$

$\Rightarrow 2Q + q_1 + 3q_2 = 0 \quad \ldots\ldots\ldots$ **(i)**

Using $V_A = V_C$

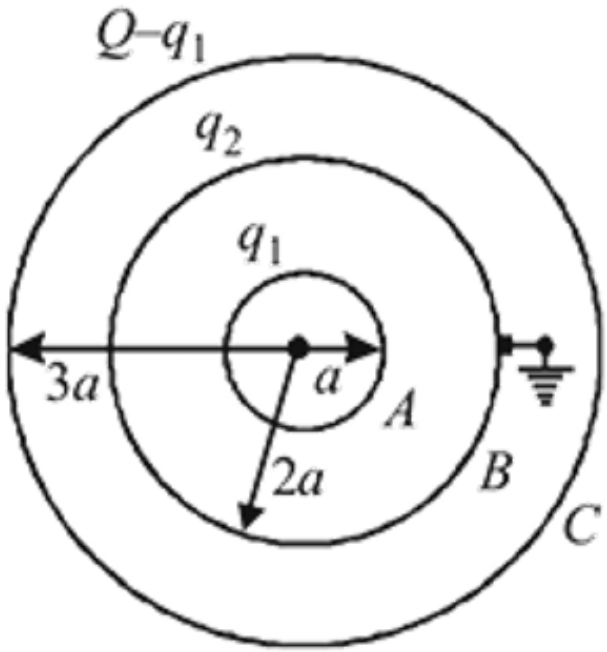
$\frac{k(Q-q_1)}{3a} + \frac{kq_2}{3a} + \frac{kq_1}{3a}$

$= \frac{kq_1}{a} + \frac{k(Q-q_1)}{3a} + \frac{kq_2}{2a}$

$\Rightarrow q_1 = -\frac{q_2}{4}$

$\ldots\ldots\ldots$ **(ii)**

Using it in $(1), q_2 = -\frac{8}{11}Q$



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Q.16 Answer:

$\frac{q \log_e 2}{4\pi\varepsilon_0 x_0}$

Solution:

$$V = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{x_0} + \frac{q}{3x_0} + \frac{q}{5x_0} + \ldots\right]$$

$$+\frac{1}{4\pi\varepsilon_0}\left[\frac{(-q)}{2x_0} + \frac{(-q)}{4x_0} + \frac{(-q)}{6x_0} + \ldots\right]$$

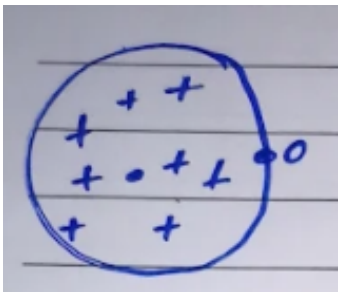
$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{x_0}\left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \ldots\right] a$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q}{x_0}\log_e(1+1) = \frac{q\log_e 2}{4\pi\varepsilon_0 x_0}$$

Q.17 Answer:

$\frac{-\rho R^2}{6\epsilon_0}$

Solution:



$V_c = \frac{3}{2}\frac{KQ}{R}$

$v_0 = \frac{kQ}{R}$

Difference

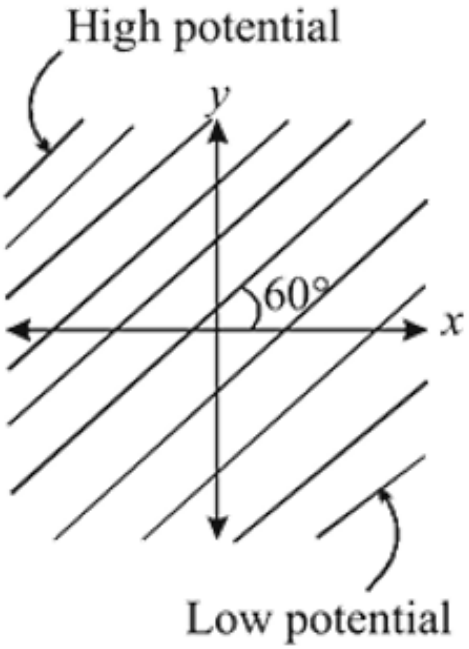
$$\Delta V = -\frac{kQ}{2R}$$

$$= \frac{-1}{4\pi\varepsilon_0} \times \frac{1}{2R}\left(\rho \times \frac{4}{3}\pi R^3\right)$$

$$= \frac{-\rho R^2}{6\varepsilon_0}$$

Q.18 Answer:

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Solution:

Electric field is perpendicular to equipotential surfaces from high to low potential

Q.19 Answer:

$\overrightarrow{E} = \hat{i}\left(2xy + z^3\right) + \hat{j}x^2 + \hat{k}3xz^2$

Solution:

The electric field at a point is equal to negative of potential gradient at that point. $\overrightarrow{E} = -\frac{\partial V}{\partial r} = \left[-\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k}\right]$

$= \left[\left(2xy + z^3\right)\hat{i} + \hat{j}x^2 + \hat{k}3xz^2\right]$

Q.20 Answer:

$-\frac{\vec{E}_0}{16}$

Solution:

$\vec{E}_0 = \frac{2kp}{r^3}\hat{p} \quad \ldots 1$

$\vec{E} = \frac{kp}{2r^3}\text{-}\hat{p}$

$\vec{E} = \text{-}\frac{1}{8}\frac{kp}{r^3}\hat{p} \Rightarrow \vec{E} = \text{-}\frac{\vec{E}_0}{16}$ **from** 1

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